Harsh Gupta

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Professional Summary

Software Engineer with experience in AI engineering, Python-based development, machine learning workflows, and AI-powered application development. Skilled in developing and evaluating AI solutions, optimizing model outputs, automating evaluation pipelines, and integrating AI capabilities into software applications. Proficient in Python, Java, JavaScript, SQL, Git, GitHub, and Docker, with a strong foundation in software engineering, data processing, debugging, and performance optimization. Passionate about building reliable, scalable, and efficient AI-driven solutions.

Education

Noida Institute of Engineering and Technology — Greater Noida, Uttar Pradesh, India
Bachelor of Technology in Artificial Intelligence; CGPA: 8.0/10 — *2021 – 2025*

Experience

Software Engineer — February 2025 – Present
Ethara AI — *Gurugram, Haryana*

- Developed and maintained Python-based tools and workflows to support AI model development, testing, and evaluation.
- Built and optimized AI evaluation pipelines to assess model outputs, identify failure patterns, and improve generated-code quality.
- Applied prompt refinement, output validation, and iterative testing techniques to improve the accuracy and consistency of AI-generated results.
- Automated data processing, testing, and evaluation tasks to improve workflow efficiency and reproducibility.
- Used Git and GitHub for version control, collaborative development, code reviews, and repository management.
- Utilized Docker to create isolated and reproducible environments for AI model execution, testing, and evaluation.
- Analyzed execution results, test outcomes, and performance metrics to identify issues and support model improvement.
- Collaborated with engineering and AI teams to troubleshoot technical issues, refine workflows, and deliver reliable AI-powered solutions.

Projects

Used-Car Price Prediction | *Python, XGBoost, Scikit-Learn, MLflow, FastAPI, Docker* — GitHub | Live

- Built an end-to-end ML service on the CarDekho dataset; benchmarked 4 regressors with 5-fold CV, and the tuned **XGBoost** model reached **R^2 = 0.921** and **MAE of INR 70K** on a held-out test set
- Packaged feature engineering, preprocessing and model in **one serialized pipeline** to prevent train/serve skew; served it via a **FastAPI** endpoint and a Streamlit UI with **SHAP** explanations; **41 pytest tests** run in GitHub Actions CI

Hospital Management System | *Java, Spring Boot, JWT, MySQL, React* — GitHub | Live

- Built a layered **REST API** in Spring Boot for patients, doctors and appointments, with a React dashboard, Flyway migrations and Testcontainers integration tests
- Implemented **JWT role-based access control** (Admin/Doctor/Patient) with ownership checks, and prevented doctor double-booking at the database level using a **generated column + unique index**

Movie Recommendation System | *Python, Scikit-Learn, FastAPI, React* — GitHub | Live

- Built a content-based recommender over **∼4,800 TMDB films** using **TF-IDF + cosine similarity**, blended with an IMDb-style weighted rating to balance relevance and quality
- Served it via a **FastAPI** backend with typo-tolerant title search (**RapidFuzz**) and a React frontend; evaluated with **Precision@k, Recall@k** and catalog coverage

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL
AI & Machine Learning: Artificial Intelligence, Machine Learning, Large Language Models (LLMs), Prompt Engineering, Model Evaluation, AI Workflow Automation
AI Engineering: Evaluation Pipelines, Output Validation, Generated-Code Quality Analysis, Data Processing
Software Development: REST APIs, FastAPI, Flask, Spring Boot, React.js, API Integration, Debugging
DevOps & Version Control: Git, GitHub, Docker, Containerization, CI/CD (GitHub Actions)
Libraries & Data Tools: NumPy, Pandas, Scikit-learn, XGBoost, MLflow
Databases: MySQL, MongoDB, SQLite
Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Software Testing, Performance Optimization

Achievements & Certifications

Naukri Campus Young Turks: Scored **99.27 percentile** in a national-level assessment
Certifications: Machine Learning with Python (IBM), Machine Learning for All (Univ. of London), Python Basics (U. Michigan)